Monitoring quality drift

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Monitoring quality drift before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Release and rollback

Without opening the old lesson, explain one key idea from Release and rollback. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Queues and asynchronous work

Without opening the old lesson, explain one key idea from Queues and asynchronous work. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Sandboxing and resource limits

Without opening the old lesson, explain one key idea from Sandboxing and resource limits. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Monitoring compares current outcomes with a declared reference and sampling protocol. A change in observed quality may reflect case mix or instrumentation.

For each window report eligible counts, passes, missing outcomes and condition mix. Alert when success falls below the toy **80%** threshold; investigate before attributing cause.

Practise with fading help

> Complete the supported Monitoring quality drift trace and record the first step you can explain without help.

Use W2: Trace the worked case for Monitoring quality **drift**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed Monitoring quality drift explanation against the supplied facts before discussing it.

Use the worked case for Monitoring quality **drift**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Window A passes 8/10; B passes 12/20. Threshold **70%**. Compare.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

Explain the mechanism. Apply the supplied Monitoring quality drift facts and keep your

> reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Monitoring quality drift facts and keep your reasoning.

Reference window passes 18/20; new window 14/20. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Monitoring quality drift facts and keep your reasoning.

A report omits this qualification: An alert identifies a condition for investigation, not a complete causal explanation. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Monitoring quality drift facts and keep your reasoning.

Window A passes 8/10; B passes 12/20. Threshold **70%**. Compare.

My reasoning and evidence

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Monitoring quality drift facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

Project checkpoint and reflection

> Connect Monitoring quality drift to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled monitoring quality **drift evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of monitoring quality **drift**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

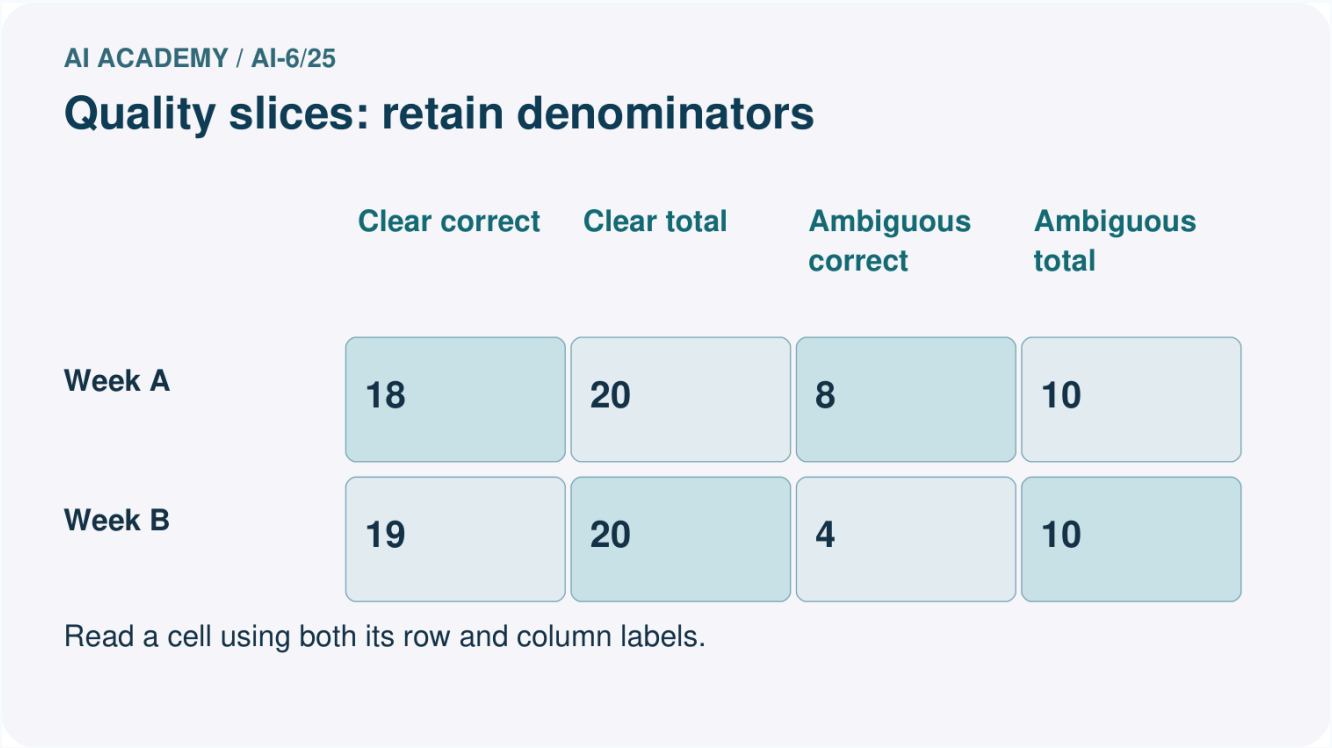
WORKBOOK · 12 OF 15

Pictorial case practice: Monitoring quality drift

> Monitor separate question slices. A service uses the same rubric on clear and ambiguous questions in two recorded weeks. Explain the labelled step before checking the changed condition.

Monitor separate question slices. A service uses the same rubric on clear and ambiguous questions in two recorded weeks.

A: clear 18/20, ambiguous 8/10. B: clear 19/20, ambiguous 4/10.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-6/25

Monitoring quality drift

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute slice rates

Clear rises 90% -> 95%; ambiguous falls 80% -> 40%.



2. Compute totals

A is 26/30; B is 23/30 correct.



3. Investigate ambiguity

Inspect that slice's failures before selecting a repair.

CHANGE ONE THING

Week B has no ambiguous questions.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Compute both clear and ambiguous slice rates for A and B. Which slice weakens, and how should that slice be reported when no ambiguous questions are observed?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Monitor separate question slices. A service uses the same rubric on clear and ambiguous questions in two recorded weeks.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Inspect that slice's failures before selecting a repair.

1. Compute totals

2. Investigate ambiguity

3. Compute slice rates

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Investigate ambiguity Inspect that slice's failures before selecting a repair. Slice monitoring exposes changes hidden by common easy questions.

Retain each denominator and compare the measurement conditions.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Week B has no ambiguous questions. Which conclusion is supported by the supplied facts?

1. That slice is unmeasured, not perfectly correct.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: That slice is unmeasured, not perfectly correct. No **observation**s means no rate can be calculated for it.

Repair a missing building block

Review Release and rollback (ai-6/24). Goal: Explain the **mechanism** and terminology of release and rollback. Short practice: Compute A's and B's correctness percentages.

Why is B still blocked, and does passing a fresh privacy check automatically satisfy every release gate?. Return to this week and explain what you changed.

Review Queues and asynchronous work (ai-6/22). Goal: Explain the **mechanism** and terminology of queues and asynchronous work. Short practice: Explain what the queued J12 submission response proves and what it does not.

Which event permits succeeded status, and what status is honest after a worker crash?. Return to this week and explain what you changed.

Review Sandboxing and resource limits (ai-6/19). Goal: Explain the **mechanism** and terminology of sandboxing and resource limits. Short practice: Check A against both resource limits and B against network isolation.

Why must A stop after three seconds even if its **output** remains only 2 MB?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
> | |
> | |
> | |
> | |

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Monitoring quality drift picture from its labels, then record a relevant limit.

01. Monitoring quality drift: the case

AI ACADEMY / AI-6/25 · 01 · LOOK AT THE FACTS

Monitoring quality drift: the case

The situation

Monitor separate question slices. A service uses the same rubric on clear and ambiguous questions in two recorded weeks.

Supplied evidence

A: clear 18/20, ambiguous 8/10. B: clear 19/20, ambiguous 4/10.

Explain monitoring quality drift using the deciding step in this fictional service.

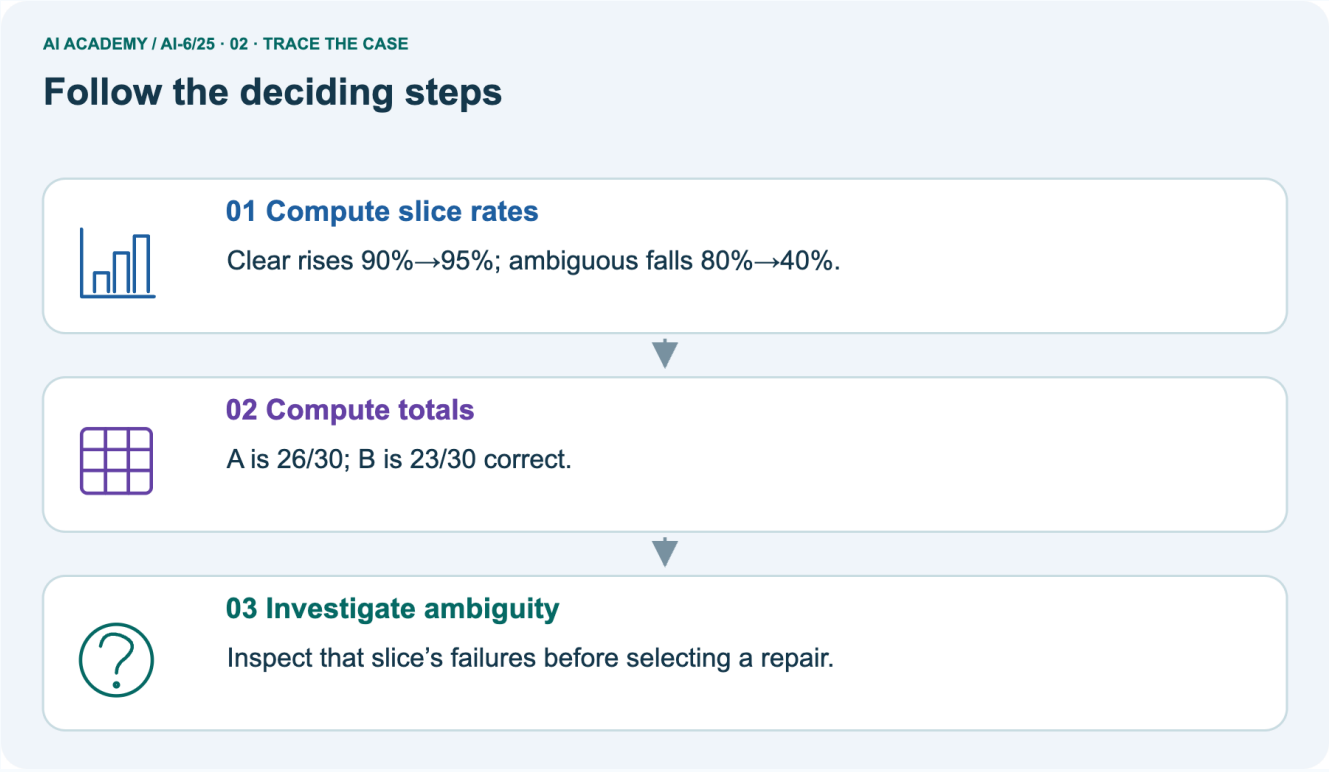
Follow the labelled picture with the matching explanation. Open the vector diagram

Monitor separate question slices. A service uses the same rubric on clear and ambiguous questions in two recorded weeks. Supplied facts: A: clear 18/20, ambiguous 8/10. B: clear 19/20, ambiguous 4/10.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



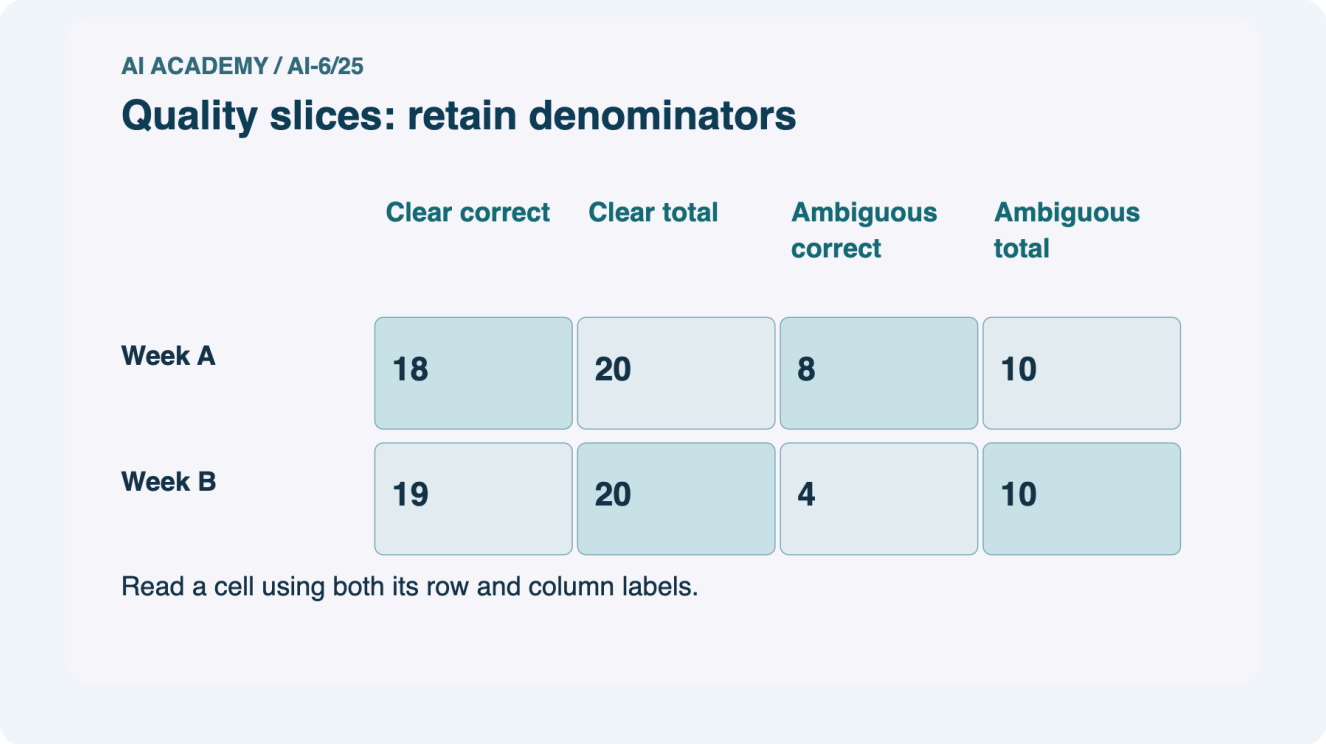
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute slice rates: Clear rises 90%→95%; ambiguous falls **80%**→40%. → Compute totals: A is 26/30; B is 23/30 correct. → Investigate ambiguity: Inspect that slice's failures before selecting a repair.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Quality slices: retain denominators



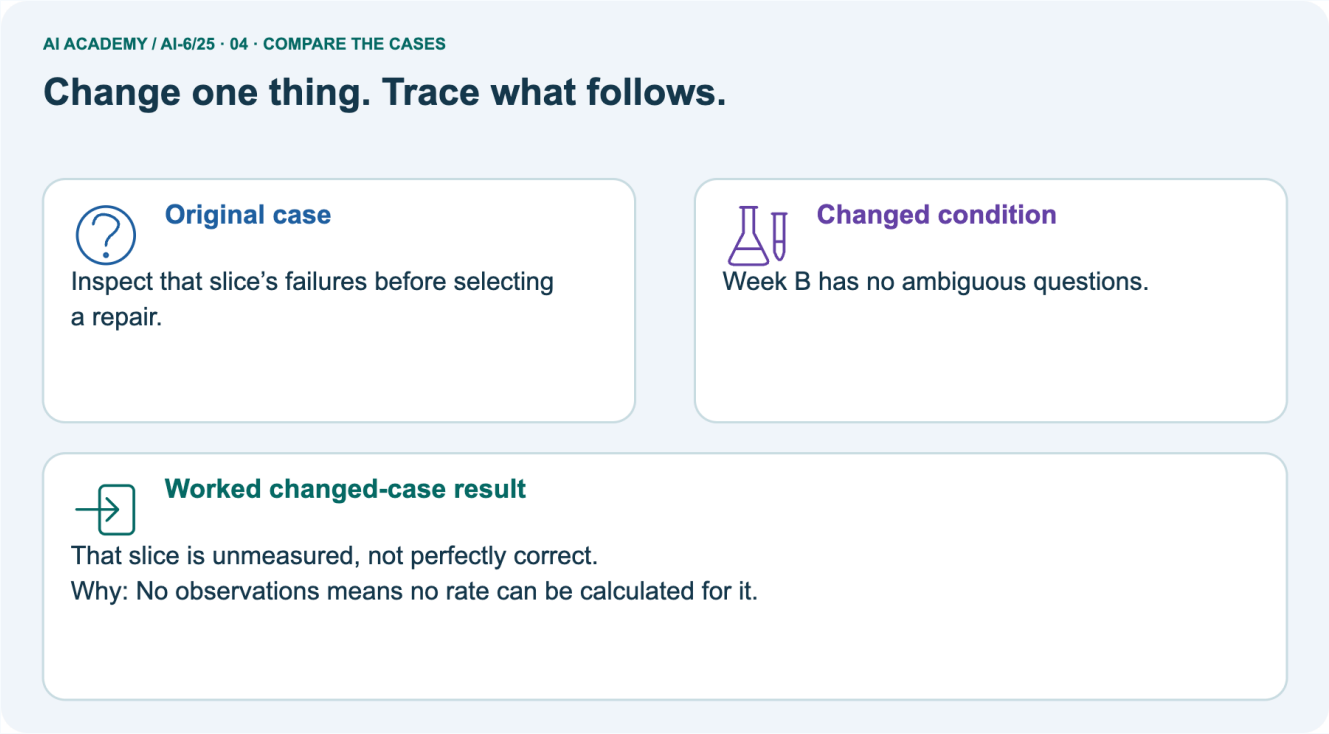
Follow the labelled picture with the matching explanation. Open the vector diagram

Quality slices: retain denominators. Week A: Clear correct = 18, Clear total = 20, Ambiguous correct = 8, Ambiguous total = 10. Week B: Clear correct = 19, Clear total = 20, Ambiguous correct = 4, Ambiguous total = 10.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

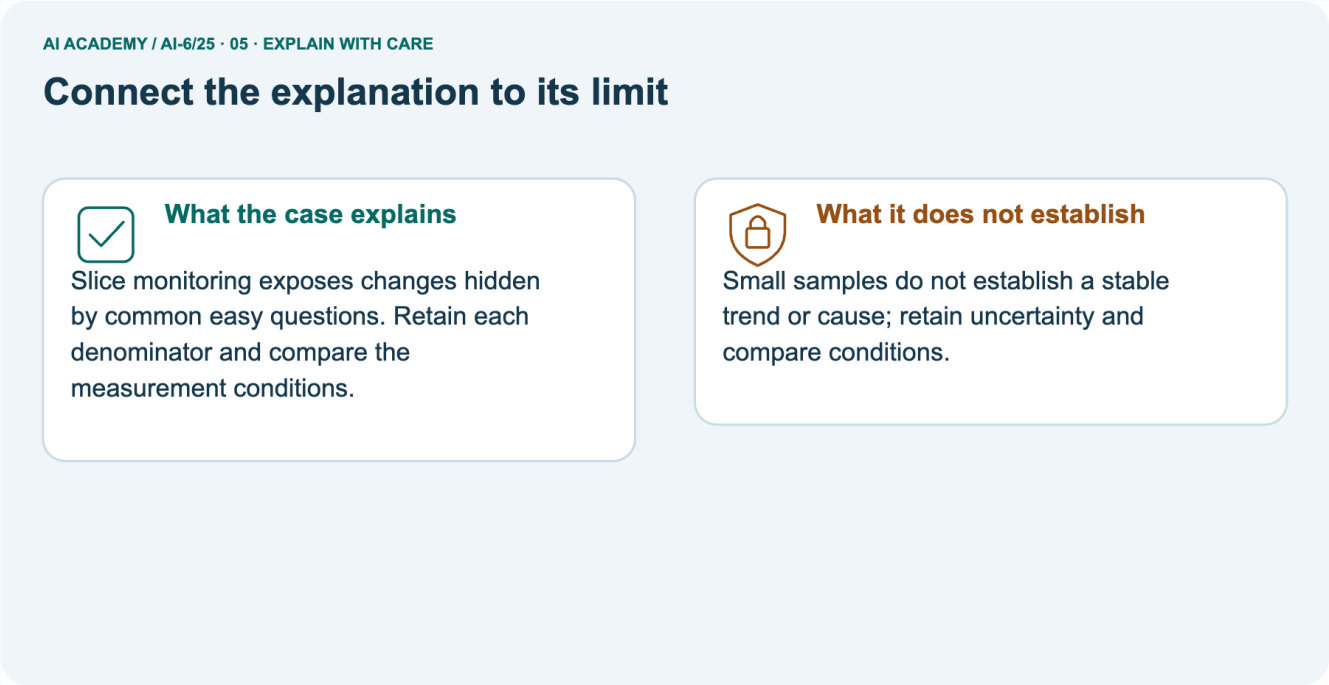
Original case: Inspect that slice's failures before selecting a repair. Changed condition: Week B has no ambiguous questions. Expected result: That slice is unmeasured, not perfectly correct.

Reason: No **observation**s means no rate can be calculated for it.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

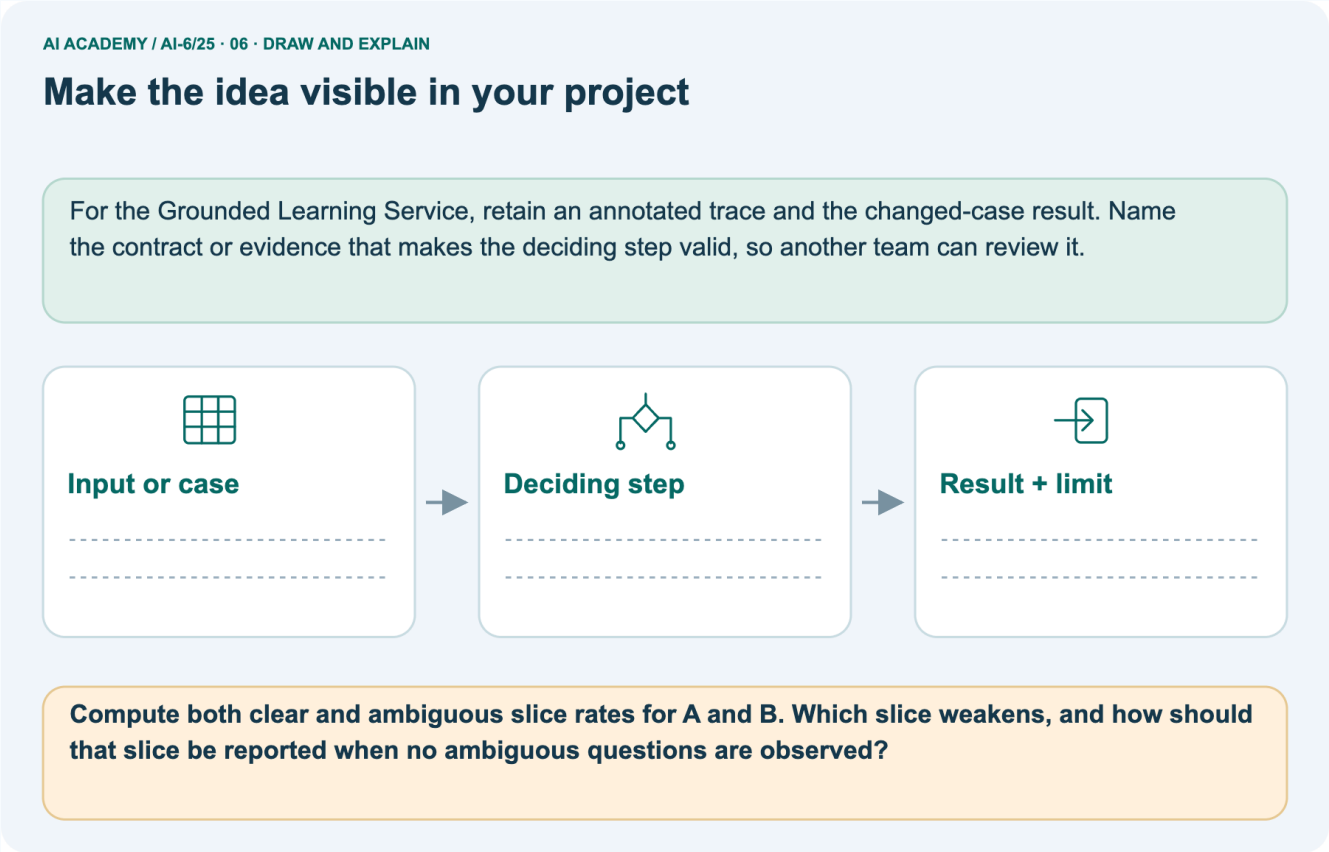
Slice monitoring exposes changes hidden by common easy questions. Retain each denominator and compare the measurement conditions.

Limit: Small samples do not establish a stable trend or cause; retain uncertainty and compare conditions.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Compute both clear and ambiguous slice rates for A and B.

Which slice weakens, and how should that slice be reported when no ambiguous questions are observed?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.